Advancements in **Wildfire Prediction** and Detection: A Systematic **Review**

V Kolluru, AN Chintakunta, Y Nuthakki, S Koganti - 2022 - papers.ssrn.com

… One big issue is the high degree of endliness of the factors that play a role in the **behavior** of fire or **wildfire**. Some factors influencing forest fires include weather conditions, physical …

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Iterated local search for the placement of **wildland fire** suppression resources

AB Mendes, FP e Alvelos - European Journal of Operational Research, 2023 - Elsevier

… **propagation** times between adjacent nodes and use the minimum travel time principle to **model** the fire **propagation** … We also provide a categorised literature **review** on fire suppression …

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A global fuel characteristic **model** and dataset for **wildfire prediction**

JR McNorton, F Di Giuseppe - Biogeosciences, 2024 - bg.copernicus.org

… The further development of a full fire risk and fire **spread model** requires consideration of ignition and **spread** based on meteorology and orography. Future modelling applications could …

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A generic, empirical-based **model** for predicting rate of fire **spread** in shrublands

WR Anderson, MG Cruz, PM Fernandes… - … of **Wildland Fire**, 2015 - CSIRO Publishing

… **prediction** errors for the **wildfire** validation datasets had no bias; this is despite the **wildfire** datasets having average rates of **spread** almost double that of the **model** development sets. …

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A **review** of approaches to estimate **wildfire** plume injection height within large-scale atmospheric chemical transport models

R Paugam, M Wooster, S Freitas… - … Chemistry and Physics, 2016 - acp.copernicus.org

… we cannot resolve the plume **dynamics**, parameterizations are … The aim of this manuscript is to **review** the different … of the Sofiev **model** against (i) the original version of the **model**, (ii) …

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Using soil moisture information to better understand and predict **wildfire** danger: a **review** of recent developments and outstanding questions

ES Krueger, MR Levi, KO Achieng… - … of **wildland fire**, 2022 - CSIRO Publishing

… **model** compared to a similar **model** without soil moisture data. … of the **wildfire** (eg rate of **spread** and fire intensity). … components describing the **wildfire** danger: **Spread** Component, …

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Using IoT and ML for **forest fire** detection, monitoring, and **prediction**: a literature **review**

M Grari, M Yandouzi, I Idrissi… - Journal of Theoretical …, 2022 - researchgate.net

… Dry vegetation fuels a **wildfire's** rapid ignition and **spread**. It … or to anticipate it, because they **spread** rapidly. Early-warning … DL to detect wildfires and their **spread** via a comprehensive …

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Fuzzy boundaries: **Simulation** and expertise in **bushfire prediction**

T Neale, D May - Social Studies of Science, 2020 - journals.sagepub.com

… ’ **prediction** (see Figure 3), is produced using a computer-based **bushfire spread** simulator. … agencies to protect themselves from ‘post-incident **review** and judgment, analysis, litigation, …

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Using spatial reinforcement learning to build forest **wildfire dynamics** models from satellite images

S Ganapathi Subramanian, M Crowley - Frontiers in ICT, 2018 - frontiersin.org

… we learn a **dynamics model** from readily available satellite image data and treating **wildfire** as an … This **review** demonstrates that while various classical AI and ML techniques have been …

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Visualization of **prediction** methods for **wildfire** modeling using CiteSpace: a bibliometric analysis

M Pan, S Zhang - Atmosphere, 2023 - mdpi.com

… It highlights the dominant areas of research, including climate change, fire **behavior**, … field of **wildfire prediction**. It facilitated a systematic **review** of the **wildfire prediction** domain through …

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